Sahaj Bhadja

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Summary

C++ software engineer with 2+ years building **high-performance autonomy tooling**, **simulation-integrated systems**, and **cross-stack production software** in fast-moving environments where correctness and latency both matter.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Designed **C++/Python service components** and onboard-style data interfaces for 500+ users, reducing end-to-end latency from **1.2s to 350ms** via profiling and runtime optimization.
- Implemented **fault-tolerant orchestration paths** for 5K+ monthly jobs, improving reliability and enabling rapid integration across distributed engineering workstreams.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **performance-sensitive product features** handling 10K+ daily requests, translating interface and execution requirements into reusable full-stack modules.
- Improved **automated release and validation** workflows in CI/CD, reducing turnaround by **60%** while maintaining stability across high-change deployments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **simulation-scale experimentation pipelines** over 100K+ trajectories, enabling reproducible benchmark execution for autonomy-oriented behavior evaluation.
- Produced **technical diagnostics and system analyses** that informed cross-functional architecture decisions and accelerated root-cause debugging.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **distributed backend services** with Python/Java and cloud data tooling, improving response times by **30%** and supporting reliable high-volume operations.
- Maintained **20+ automated test suites** across Linux/Windows with Jenkins, PyTest, and Mockito, improving pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket workflows** for eight modules and 2M+ synchronized records, improving concurrency handling and end-user responsiveness.
- Automated **containerized deployment pipelines** with Docker/Jenkins, reducing release time by **50%** and improving repeatability of production rollouts.

Skills

Languages: **C++11/14/17**, **Python**, Java, JavaScript/TypeScript, SQL, Bash
Autonomy & Simulation: virtual scenario testing, synthetic data workflows, real-time behavior validation
Systems: concurrency, multithreading, performance profiling, Linux/UNIX development
Infrastructure: distributed services, CI/CD automation, Docker, Kubernetes basics, Git/GitHub
Engineering: interface design, cross-stack collaboration, architecture reviews, production troubleshooting

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLFQ algorithms** with metrics and visualization, reducing debugging time by **20%** through deterministic, replayable scenarios.

Autonomy Scenario Replay Tool | *C++, Python, JSON, OpenGL*

- Built a **high-performance replay pipeline** for sensor/event streams with configurable timing and contract checks, improving integration test confidence for mission software.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized REST processing service** with automated CI/CD validation, improving reliability and operational continuity for production-grade workloads.